RAABX — Roll-ups As A Business Whitepaper

Bitcoin-Cadence L2 with Instant/Cut-off Seals, zk-KYC, and UDS

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Abstract

RAABX is a Bitcoin-anchored Layer-2 backbone for business that cleanly separates a Public (zk-KYC) lane from a Private (confidential) lane while preserving Bitcoin-level auditability and predictable cost. Blocks follow a 10-minute cadence; Instant Seal publishes the current block in ~2 s when latency matters, and Cut-off Seal closes named scopes (e.g., a payroll run) inside an open block—so operators control when things settle without ever changing the anchor schedule. Consensus is stake-weighted Nakamoto on a BTC-style fork with VRF leadership; Instant Sealed blocks finalize by rule via SealCert. Identity uses zero-PII zk-proofs with fast revocation; accounts use hybrid ECDSA + PQC keys. RBS is the fee/staking unit, and UDS auto-delegates balances so every holder earns by default, with health-weighted rewards and slashing for faults. Anchors post to Bitcoin twice daily, funded by an Anchor Pot sized for runway, yielding fixed-time, Taproot-verifiable receipts. A single gateway origin serves wallet, RPC, sealing, explorers, and anchor lookups. In short: explicit timing, private-by-design compliance, and an always-earning economic base—engineered so businesses can reason about finality, prove it to others, and operate at internet speed. Sampling uses VRF-weighted selection without replacement; a reveal phase prevents precomputation and resists bribery/censorship replay.

Document Conventions & Time Definitions

• Block (a “payout epoch”) — a sealed block. Rewards stream per sealed block.

• UDS shuffle — a daily reassignment window (~24 h) where UDS rebalances via VRF. Separate from block sealing.

• Anchor slot — fixed, twice-daily Taproot posts to Bitcoin (time-boxed windows; seals never trigger anchors).

Where this document says “epoch,” it means block/payout epoch unless explicitly called “UDS shuffle.”

Approximation style: we use ~ (tilde) for approximations (e.g., ~2 s).

RAABX Protocol: Bitcoin-Cadence Backbone for Business

RAABX is a Bitcoin-anchored L2 backbone with two parallel lanes: a Public “KYC” (Know Your Customer) lane and a Private “confidential” lane. It targets business-grade UX, strict public/private separation, deterministic finality when sealed, fixed-cadence L1 anchoring, its own Signet, and crypto-agile security.

Organizations can add paired L3s (one public, one private) that inherit L2 security. The Public lane handles customer flows (payments/DEX with zk-KYC) while the Private lane handles payroll/contracts with shielded accounting and StorageReceipts. Orgs can sponsor gas or set zero internal fees.

Blocks close on a fixed 10-minute cadence (Auto Seal). Anyone may request an Instant Seal to close and publish the in-progress block in ~2 s (fee + rate limits apply); exactly one Instant Seal may succeed per block, the earliest valid request must be included, and the block is final once a sampled stake committee issues a SealCert. For scoped business events, a Cut-off Seal finalizes a named accounting scope inside the still-open block—freezing that scope while other activity continues—and commits its ScopeRoot into the block roots. If neither occurs, longest-chain finality applies at the 10-minute close. Seals govern ledger timing only; Bitcoin anchoring runs on fixed twice-daily UTC slots. Consensus is stake-weighted Nakamoto on a BTC-style fork with a VRF (Verifiable Random Function) leader. Instant Sealed blocks become final via a sampled SealCert (supermajority of sampled stake); otherwise longest-chain rules apply. Inclusion rules, rate limits, and slashing punish equivocation or censorship.

All important events anchor to Bitcoin twice daily at fixed UTC slots with slight smoothing. An Anchor Pot, funded largely by Instant Seal fees, is managed to keep ~30 days of runway; provenance flows ScopeRoot → ModuleRoot → BlockRoot (chain ID, height, receipts root, module roots, params hash, time range) → Bitcoin Taproot.

Accounts use unified addresses and hybrid keys (ECDSA + PQ signatures like Dilithium/SPHINCS+), with optional thresholds, time-locked rotation, and KYC-assisted recovery. Transport uses QUIC+Noise; Private-lane payloads can be encrypted; anchor receipts can include PQ co-signatures.

Network/MEV hygiene includes refundable anti-spam deposits, deterministic inclusion order, priority lanes (Instant Seal, scope-close), per-key seal rate limits, and an explicit inclusion rule to reduce frontrunning and censorship.

RBS is the unit for both fees and staking, and by design every RBS balance is always earning. The protocol runs UDS auto-delegation (Universal auto-Delegation & Staking), which automatically assigns your balance to a validator—no pool picking required. At each UDS shuffle (~24 h), a VRF reassigns balances toward validators with the best combined stake^β × health score; rewards auto-compound by default (toggle in-wallet). Validators that underperform are auto-unbonded, while slashing covers faults like equivocation or breaking the inclusion rule. This 'always-earning' model secures the chain without user micromanagement, while fee flows and Instant Seal fees fund the Anchor/Perma pot and the DAO per policy. (default β = 0.5; governance envelope 0.3–0.8) Health is evaluated over a rolling 24-hour window to avoid thrash.

Public data is receipt-level only (no PII); Private data is shielded with auditor keys. RAAB-Signet (minimum of 3-of-5 HSM signers, ~10-minute blocks) lets teams test end-to-end sealing and anchoring. A single gateway origin serves the wallet, public/private RPC, /seal, explorers, and anchor lookups.

Threat drills focus on concrete failure modes—seal-request spam, Anchor-Pot depletion, censorship/MEV, stale revocation roots, privacy leakage, and key compromise—and drive our operational defaults: blocks close every 10 minutes with an optional Instant Seal; the Instant Seal fee floor is adaptive (it rises under load and is sized to preserve the Anchor-Pot runway); anchors to Bitcoin occur at fixed twice-daily UTC slots; validators are provisioned at roughly 8 vCPU / 32 GB RAM / NVMe / 1 Gbps with replicated RPC gateways; and we expose first-class telemetry—instant_seals (rate), seal_floor (current minimum), anchor_pot (balance/runway), anchor_posted (last anchor status), and scope_closes (cut-off seals). We publish implementation notes for seal handlers, BlockRoot computation, the anchor aggregator, and explorer proof drill-downs so behavior is auditable end-to-end.

Security posture. We use hybrid signatures (ECDSA today with Dilithium/SPHINCS+ ready) under an add-only cryptography governance model: we may add algorithms or rotate keys over time, but do not remove previously accepted schemes. Anchor receipts can optionally include post-quantum co-signatures, providing forward-looking assurances without breaking compatibility.

Taken together, RAABX couples Bitcoin-cadence blocks and twice-daily anchors with immediate, deterministic sealing (Instant/Cut-off), zero-PII zk-KYC, and an 'always-earning' staking base (UDS). The result is a backbone that preserves privacy by construction,

produces audit-grade receipts on a fixed schedule, and keeps operational costs predictable through the Anchor Pot. The remainder of this paper specifies consensus and sealing, zk-KYC and identity, token supply and rewards, operations and threat drills, and governance—each with explicit acceptance criteria and anchor-verifiable proof paths.

Zero-PII Onboarding (zk-KYC)

What it is & goals. Prove compliance without ever putting personal data on-chain. Verify once, reuse proofs; instant revocation; policy-flexible predicates (age, residency, sanctions, accreditation, one-per-human); crypto-agile (hybrid PQC).

Actors & flow. In-wallet, a user completes KYC with two independent oracles (off-chain). The wallet stores their signed credentials locally and mints a non-transferable KYC-PASS on the Public L2 that holds only a commitment hash. Public & Private wallets are created; every gated action includes a fresh, unlinkable zero-knowledge proof. Urgent actions can finalize via Instant Seal; otherwise they finalize on the 10-minute Auto Seal cadence.

Data & proving. Device-held credentials contain issuer IDs, schema/cred IDs, expiry, attribute commitments, a link-secret, and the issuer's signature. The on-chain KYC-PASS stores only commitments (no PII). Each proof shows: two allow-listed credentials, same link-secret, unexpired and not revoked against the latest on-chain roots, and that required predicates hold; an optional scoped nullifier supports "one-per-human" within a defined scope. Proofs are PLONK-based (browser-generated), verified at constant cost on-chain, and can be batched.

Revocation & policy. Two independent providers publish revocation-tree roots every **≤30 s**. A zk-KYC proof is **valid only if** the referenced CRL root is **≤120 s** old at verification time.
Outage behavior (no-grace by default). If **both** CRL feeds are older than **120 s**, **KYC-gated transactions are invalid** and must be retried; non-KYC flows continue. Finalized blocks remain valid (no retroactive effects).
Emergency override (bounded). Governance may raise CRL_MAX_AGE up to **900 s** for **≤24 h** via an on-chain emergency action (15-minute timelock, 3-of-5 multisig). The override is auditable and auto-reverts to **120 s** when it expires.

Privacy & UX. Observers see only "valid proof for Policy X" and fees/scope tags; identity, attributes, and chosen issuers remain hidden. No cross-tx linkage; Public-lane proofs cannot be linked to Private-lane activity. Proofs typically generate in sub-seconds to a few seconds; gas can be sponsored for $0 UX.

Trust, governance, and costs. Oracles are allow-listed and HSM-backed; governance can add algorithms (not remove), and can penalize or remove stale/misbehaving oracles. Issuers charge for initial issuance/refresh; daily use is free (proofs are reusable) aside from tiny verification gas.

Security model. Mitigations cover credential forgery (issuer allow-list/signature checks), duplicate participation (scoped nullifiers), oracle collusion (two-oracle rule + governance), revocation lag (freshness bound), linkability (fresh proofs + secret separation), PII leaks (commitments only), and replay (binding to action hash/epoch).

Anchoring & implementation. Revocation roots, policy updates, and PASS lifecycle events land on the Public L2, commit into BlockRoots, and anchor to Bitcoin twice daily. Modules (passport, policy, rev-roots, zk-verify) and PLONK circuits (BASE/AGE18/RESIDENCY/ACCREDITED/SALE_ONCE) ship with explorers that surface CRL freshness and anchor receipts.

RAABX Passport & Profile (MVP)

What it is. A privacy-preserving identity layer: a non-transferable Passport (SBNFT) on the Public L2, an encrypted off-chain profile with on-chain StorageReceipts on the Private L2, a minimal Public Profile Card for Commons, and a RAABX Mail address bound to the Passport. Writes land in the current block (Auto Seal ~10 min or Instant Seal now) and anchor to Bitcoin on a fixed twice-daily schedule; seals never trigger anchors.

Design & onboarding. No PII ever touches chain: Public holds commitments/events; Private holds integrity receipts; ciphertext lives in BYO or Spur storage. Secrets for profile and mail are independent of zk-KYC. Flow: pass zk-KYC → mint Passport with an initial profile root → claim a mail handle and publish a mailbox key → add encrypted profile data → upload ciphertext (get a StorageReceipt) → update the Passport root; finalize on block close, then anchor on schedule.

Data model. Passport SBNFT: {passport_id, profile_root, flags, joined_at, sig_policy} (soulbound; owner can only update the root). Encrypted Profile Tree: salted, hashed leaves (work/edu/skills/bio/links/portfolio). StorageReceipt (Private L2): {cid, size, mime, retention, checksum, provider}. Expired receipts indicate blobs scheduled for purge; DSAR/erasure requests rotate or delete ciphertext off-chain while on-chain receipts remain as non-PII integrity records. Mailbox mapping (Public L2): {handle, mail_addr, pk_mail, owner_passport_id, aliases[]} (non-transferable; key rotation allowed).

Operations. MintPassport, ClaimMailbox / RotateMailboxKey, PutStorageReceipt, UpdatePassportRoot. All finalize on Auto/Instant Seal and anchor 2×/day; explorers show txids and Merkle paths.

Security & privacy. Zero PII on-chain; unlinkable proofs; client-side encryption; tamper-evident blobs; E2E mail; hybrid keys (ECDSA + PQC) with clean rotation/recovery; governance can add algorithms but not remove.

Availability & recovery. Encrypted backups for profile and mailbox keys; root can be recomputed on restore; account recovery supports time-locked rotation and zk-KYC-assisted flows.

Governance & presentation. Network-wide @handle names with anti-squatting rules; optional Public Profile Card (badge + handle + optional avatar) that never exposes the profile root.

Interfaces & observability. Gateway endpoints for profile upload/download and mail claims; explorers for Passport/Mailbox/StorageReceipts with anchor proofs; telemetry for passport/mailer events and anchor status.

Out of scope (MVP). Issuer attestations on profile claims, hiring predicate proofs, SMTP federation, and marketplace features (planned later).

Acceptance, at a glance. After zk-KYC, users can mint a Passport, claim a mail address, save encrypted profile data with receipts, rotate keys, and see anchor proofs in explorers on the fixed cadence.

RAABX Commons (MVP) — Secure, in-wallet collaboration

What it is. Identity-aware collaboration built into the wallet: public rooms, org rooms, private groups, DMs, voice/video ("Commons Live"), and RAABX Mail. Private spaces are end-to-end encrypted; on-chain receipts prove who/what/when without ever storing content or PII.

How it works. A lightweight homeserver stack ("RAABX Relay") handles auth (sign-in with Passport), chat/file pinning, media, and mail. Messages, media, and emails live off-chain in BYO storage or Spur; the chain records Chat/Call/Mail receipts on the Public lane and StorageReceipts on the Private lane. Blocks finalize on Auto Seal (~10 min) or Instant Seal (now), and everything anchors to Bitcoin on a fixed twice-daily schedule.

Spaces & sessions. Channels include public, org public, private, bulletin (read-only), and DMs, plus Live calls: peer-to-peer mesh for small groups or SFU (Spur/BYO) for larger calls—both E2E. Participants must hold a Passport; orgs moderate their own public rooms; private rooms are invite-only.

Security & privacy. Content is client-encrypted (X25519 + AEAD); keys never touch chain. Receipts prove order, authorship, and integrity (no plaintext). Optional "Channel Cut-off" fences a transcript for audits without closing the block. Transparent moderation/policy changes emit on-chain events and inherit Bitcoin anchors.

User experience. Everything runs inside the wallet: room list, chat, participants, quick actions, anchor countdown, and explorer links to inclusion/anchor proofs. Mail is E2E in the same UI; attachments are encrypted blobs with receipts.

Business Launchpad — Your Company as a Layer-3 (L3)

What it is. One-click, org-sovereign L3 with twin ledgers—Public (customer) and Private (confidential)—branded to you, policy-controlled internal gas (you can sponsor or charge, including min gas = 0 for $0 UX), and Bitcoin-anchored twice daily. Instant Seal closes & publishes your current block now (~2 s); Cut-off Seal freezes a named scope (e.g., PAYROLL:RUN#57) inside the still-open block. Seals never trigger anchors.

How layers commit. Your L3 scopes → L3 BlockRoot (Auto Seal ~10 min or Instant) → L2 AnchorTree → Bitcoin Taproot 2×/day (±15 min smoothing). Optional emergency direct-anchor from L3 to Bitcoin.

What Launchpad deploys. Twin ledgers; org website (Pages) + Storefront/POS; Commons (chat/DMs/Live) and Mail (E2E); DEX; public/admin explorers with anchor proofs; Dev Studio; full branding & theming.

Back-office engines. Inventory & POS, Cash & Treasury, OpenTax (tax archetypes), AR/AP, Payroll & HR, Procurement, Contracts & Signatures, Reporting & export (GL/IFRS/GAAP drill-downs).

Fees & finality. Inside L3: sponsor or set min gas = 0; Instant Seal can be free or policy-priced; Cut-off is a normal tx. Fees appear mainly on cross-layer/cross-org moves or managed Spur services.

Split Graph finance. Every sale auto-splits to taxes, COGS reserve, royalties/commissions, optional payroll accruals, fees, and profit—posting public aggregates vs private details correctly and inheriting the Bitcoin anchors.

Front office. Website + checkout, QR POS, receipts/warranties as content-addressed docs with StorageReceipts; customer explorers show Bitcoin-anchored proofs.

Bootstrap flow (~10 min). Claim name/domain → choose BYO or Spur infra → import OpenTax packs → map tokens↔GL → add SKUs/prices → brand/theme → clone Commons/Mail/DEX/Explorers/Dev Studio → go live.

Demos (at a glance). Retail “Candy Bar” sale: tax/COGS/royalty/payroll accrual split, instant or timed finality, auto-reorder via PO/GRN, anchor path to Bitcoin. Payroll: cents-per-sale accruals fund pay runs; run weekly/biweekly with optional Cut-off Seal; stubs/receipts anchor on schedule.

Security & compliance. Zero PII on-chain; E2E chat/mail/live; hybrid ECDSA+PQC keys; clear Public vs Private boundaries; OpenTax compliance; transparent, anchor-visible policy/moderation events.

Observability & cost. In-wallet explorers and an Anchor Panel (countdown + last txids + Merkle paths). Blocks ~10 min by default; anchoring batched 2×/day from the shared

Anchor Pot; QUIC networking; predictable costs.

RAABX Dev Studio — The Business Builder (No-Code → Pro-Code)

What it is. The control room to move your business onto RAABX: stand up your org's Public + Private L3s, configure finance/ops, and ship websites/apps—all inside the wallet. BYO infra or managed Spur; every blob gets a StorageReceipt, every deploy emits a signed manifest, and anchors land on Bitcoin 2×/day (seals never trigger anchors).

Who it's for. Operators use guided wizards and templates; builders drop into embedded IDEs, Covenant Lab, ZK Proof Lab, and a Tx Builder/Inspector for custom logic.

Core surfaces. Launchpad Wizards (OpenTax, Inventory/POS, Cash/Treasury, AR/AP, Payroll/HR, Procurement, Contracts, Reporting, Websites/Apps), Templates Library, Website options (keep, clone to RAABX Pages, or build new), Ops Console (roles, budgets, logs, Anchor Monitor), and one-click hosting with BYO/Spur for storage/compute/SFU.

Timing model. Auto Seal closes ~10-min blocks by default; Instant Seal publishes now (~2 s) when you need it; Cut-off Seal freezes just a business scope (e.g., a payroll run) inside the still-open block. Anchors are fixed-cadence Taproot posts 2×/day.

Security & privacy. Keys stay local; zero PII on-chain; content is client-encrypted off-chain with StorageReceipts on the Private L3; policy changes and deploys leave verifiable on-chain receipts.

Day-1 to live. Create org → install blueprints → import catalog & tokenize SKUs → choose website path (embed/clone/build) → connect payments & org-sponsored gas → test (Cut-off a shift, Instant Seal a block) → go live and watch anchor proofs in explorers.

MVP → Phase-2. MVP ships wizards/templates, checkout widgets, OpenTax + Split Graph, Inventory/POS, Payroll/HR, Contracts, Reporting, Commons/Mail, hosting, DevPods, Covenant/ZK labs, Tx tools, faucets, and anchor/audit tooling. Phase-2 adds GPU DevPods, a template marketplace, deeper tracing/migrations, connector galleries, and expanded covenant/policy catalogs.

Token Library, OpenTax & Cross-Org Liquidity

RAABX ships a production-grade set of business primitives that work the same on every org's twin L3s (Public + Private), settle via L2, and anchor to Bitcoin twice daily. Core

standards—IRaab20 (fungible), IRaab721 (NFT/SBNFT), and IRaabFeed (time-series)—have a canonical RBS-native reference implementation, zero PII on-chain, encrypted blobs via StorageReceipts, and first-class Instant and Cut-off Seals.

The token library covers retail (SKUs, price lists, offers/loyalty, gift cards, inventory, orders/shipments/returns, warranties, lots/serials, tenders, store sessions), payroll & HR (employee records, terms, shifts, pay stubs, payroll batches, withholdings, employer liabilities, benefits, 401k, garnishments, union dues, PTO, expenses), supplier programs (allowances/rebates/chargebacks, VMI), and logistics (multi-DC, transport legs, recalls).

IRaab20 supports policy gates, role caps, org scopes, optional GL maps, and deterministic Split-Graph postings; IRaab721 adds soulbound flag, content-addressed metadata, and scope tags for clean Cut-off sealing; IRaabFeed provides append-only telemetry with retention/compaction and paginated proofs.

Sales/Inventory/Logistics objects are public when they're receipts and private when sensitive. Restricted tenders (SNAP/WIC/EBT) are enforced with predicates (no PII), and tips/service charges are split correctly (including FICA tip credit capture).

Split-Graphs map each sale to cash legs (merchant, taxes by jurisdiction, royalties/allowances, loyalty, tips, fees) and GL legs (revenue, COGS, inventory, payroll accruals, liabilities/deferrals), with compliance gates (e.g., zk-age), multi-state apportionment, and Cut-off Seals to freeze POS shifts or invoice cycles.

OpenTax v1.1 includes deep US payroll & sales/use coverage plus EU/UK/AU/CA packs, with quarterly/annual filings and payment artifacts (ACH/SEPA) hashed/anchored while PDFs remain encrypted off-chain.

The Payroll Engine (Private L3 → Public aggregates) consumes shifts, terms, benefits, garnishments, 401k elections, and Split-Graph accruals; computes gross → pre-tax → taxes → post-tax → employer taxes; mints private PayStubReceipts, produces public PayrollBatch, and seals each run (Cut-off, optionally Instant). Edge cases (multi-state, reciprocity, supplemental rates, mid-period moves, PTO payouts, final checks, minimum-wage top-ups) are handled.

Benefits/Insurance/Compliance include 401k/403b/Roth (limits, catch-up, vesting, loans metadata), HSA/FSA/commuter, medical/dental/vision/life/STD/LTD splits, ACA thresholds/1095-C, Workers' Comp buckets/rates, SUTA experience, COBRA notices, I-9/W-4 refs, and expiring training/cert tokens with role gates.

Logistics & Cold-Chain track warehouses/DCs, exceptions (short/over/damaged), GPS/locality proofs, SLA credits/chargebacks, and sensor telemetry; breaches can trigger automatic recalls/refunds/chargebacks with anchor-visible audit trails.

Post-MVP Router enablement targets cross-org liquidity (factoring invoices, dynamic discounting, securitizing royalties, carbon bundles) with atomic L3↔L3 settlement via L2 (HTLC/adapter proofs), plus risk guards (whitelists, price bands, circuit breakers, zk-KYC).

Privacy & Anchors: public L3 exposes commitments/events (no PII); private L3 stores encrypted docs with residency tags; both L3s roll into L2 and anchor to Bitcoin twice daily; hybrid signatures are supported.

Interfaces include public business events (sales, tenders, payroll, tax, POs/ASNs/GRNs, supplier settlements, returns, rebates/chargebacks, shipments/recalls, policy changes, exports, scope-close) and private receipts (StorageReceipts, pay stubs, COBRA, I-9/W-4).

An acceptance suite proves end-to-end correctness (multistate payroll, SNAP baskets, mixed baskets with coupons/loyalty/age-gates, tip handling, mid-year benefits changes, garnishment precedence, Workers' Comp audits, returns/refunds, cold-chain breaches, and quarter close). MVP ships the full retail set, core HR/payroll objects, Split-Graphs, OpenTax US + base international packs, GL postings, offline POS, StorageReceipts, exports, anchors, explorers, and seal UX; fast-follow adds ACA deep pack/COBRA flows, Workers' Comp UI, vendor scorecards, VMI, advanced FX/hedging, cold-chain dashboards, Router enablement, curated predicate packs, and international payroll.

Router & Liquidity (Cross-Org DEX, Tenders, Escrows)

RAABX's Router lets organizations exchange value and run commercial workflows across their own L3s—no bridges or custodians. Deals are two coordinated legs (lock on the source, release on the destination) notarized at L2; if anything fails or times out, funds refund. Policy is enforced at execution with zk-KYC/KYB and OpenTax tags. Timing stays predictable: use Cut-off Seals to fence bidding/auction scopes and Instant Seals when you need the block to publish now; anchors to Bitcoin remain on a fixed twice-daily schedule.

Atomicity & Timelocks

We execute cross-rail transfers using **hash-locked** flows (HTLCs or adapter-signature equivalents) so swaps either **complete atomically** or **refund**.

Defaults (safe for today's ≤3 s Instant finality):

- T_src (source timelock): **1800 s** (30 min)
- Δ (safety delta between rails): **600 s** (10 min)
- T_dst (destination timelock): T_src − Δ = **1200 s** (20 min)
- skew_budget: **≤500 ms** tolerated clock skew
- e2e_p99: measured end-to-end settlement (target: **≤3 s**)

Rules (copy-paste into implementations):

1. **Choose Δ conservatively:** Δ ≥ 3 × e2e_p99 + network_jitter_margin (the **10 min** default covers this comfortably).
2. **Refund ordering:** if the destination claim hasn't landed by T_dst, the **destination leg refunds first**; the **source leg refunds** after T_src.
3. **Preimage propagation:** once revealed on any leg, the counter-leg can be claimed immediately; operators must monitor both legs continuously.
4. **Bitcoin hops:** when a BTC L1 leg is involved, set the CLTV delta to cover **≥2 blocks plus** Δ.
5. **Multi-hop:** timelocks **decrease** from source → destination so earlier legs always have **more time** than later legs.
6. **Interop knobs:** hash = **SHA-256**; time units = **seconds** on L2, **blocks** on BTC L1.

These parameters are intentionally conservative and can be tightened later with live e2e_p99 telemetry.

An L2 registry defines each asset (class/decimals/supply flags) and its compliance predicates (e.g., KYC/KYB, residency, accreditation, region allow/deny). The Router checks these rules with fresh revocation roots before executing.

Three protocol families cover most flows. XSwap moves value for value in two ways: (a) pure HTLCs for permissionless atomicity, or (b) Router-quoted RFQ where makers return prices/paths and an L2 SwapCommit notarizes the route; L3 emits EscrowLock/Release/Refund receipts. XTender runs sealed-bid procurement and job bids with commit-reveal (optional encrypted bid payloads), zk eligibility checks, deterministic or committee awards, and an AwardEscrow for milestones/deliverables. XEscrow is the general-purpose escrow for milestones, closings, and SLAs across domains like real estate, construction, and SaaS. All writes land in L3 receipts and anchor to Bitcoin on the fixed cadence.

RAABX BusinessNet (Runtime, Discovery & Audit)

BusinessNet is the in-wallet runtime that takes what you publish in Dev Studio live: it serves your org's sites and apps, enforces release integrity, handles identity/access, drives discovery/ranking, and exposes audit trails—all without leaving RAABX. You can host on

your own infra (BYO) or use Spur's managed SLA, and switching providers is just a config flip.

How it serves: the wallet talks to a single Pages Gateway that serves static assets, proxies your DevPod APIs, and refuses anything not in the signed Site Manifest. Assets are content-addressed (CIDs) with StorageReceipts; each deploy posts a manifest to the Public L3 and gets anchored to Bitcoin. Domains bind via DNS + a well-known manifest fingerprint; "Sign in with RAABX" gates routes by badges, roles, or zk-predicates (no PII on-chain). The edge runs HTTP/3 with immutable caching and per-org isolation. Seals are available at runtime: Cut-off to freeze a scope mid-block, Instant for ~now settlement, or default Auto (~10 min).

What you can run: static sites; dynamic micro-services (REST/GraphQL/indexers/webhooks/cron); Commons Live (P2P Mesh for small, SFU for scale with E2E media and optional encrypted recording); and built-in payments/proofs with OpenTax and optional zk-checks.

Discovery & operations: orgs publish manifests to the BusinessNet Index; ranking uses relevance, trust, performance, freshness, and uptime. Sponsored slots are opt-in, clearly labeled, and leave on-chain AdReceipts (revenue splits per the Economics: Fees, UDS Staking, Anchor Pot & Revenue section). Choose BYO (sovereign control) or Spur (SLA + on-chain downtime credits), and switch via a single config. Security/compliance: signed releases + strict CSP nonces, end-to-end crypto for private mail/live, keys never on-chain, right-to-erasure at the source, and full proof paths (manifest txid, asset CIDs, Scope/BlockRoot, Merkle path to Bitcoin) for audits.

Performance, Sizing & Limits

UX timing is fixed by seals and anchors: Auto Seal closes blocks on a ~10-minute cadence (avg wait ~5 min); Instant Seal publishes the current block in ~2 s; Cut-off Seal freezes a named business scope without stopping the block. Anchors post to Bitcoin 2×/day with a ±15-minute fee window—seals never trigger anchors. Canonical windows: 00:00 and 12:00 UTC ±15 min.

Throughput targets. L2 sustains ~**150–300 tx/s** with bursts ~**500 tx/s**. Each org's Public/Private L3 targets **~50–150 tx/s**, and capacity scales horizontally across many org L3s.

Sealing economics. Exactly one Instant Seal can succeed per block; the earliest valid must be included. The adaptive floor is minSeal = max(Fmin, α·p95Fee)·(1+δ). Governance maintains a ≥30-day Anchor-Pot runway; δ nudges the floor up if runway dips (0 when runway ≥30 d; 0.25 for 14–30 d; 0.5 for <14 d; warn at <30 days, page at <14 days).

Example: if p95Fee = 40 sat/vB, Fmin = 35 sat/vB, α = 1.0, and runway tier δ = 0.25, then minSeal = max(35, 40) × (1 + 0.25) = 50 sat/vB equivalent; the wallet quotes the actual Instant Seal fee in RBS using current BTC/RBS rates and the byte-size of the commitment. Quotes use a median of L2 DEX TWAPs over 30–60 minutes, optionally cross-checked by an external oracle; the floor logic uses the BTC L1 p95 fee oracle (sat/vB), converted to RBS/byte; if the oracle is stale, the policy floor Fmin applies..

Anchoring strategy. The protocol posts exactly **two** anchors per UTC day at **00:00** and **12:00**. If a slot is **missed**, it is **not** made up; the **next scheduled slot** commits **all** unanchored AnchorTree leaves up to that time. Over any rolling **24 h** window the target is **≥2** anchors (may drop to **1** only if **consecutive** misses occur). Operational alerts fire on any miss; consensus rules are unchanged.

Fees & UX (org-controlled). Inside an org's L3s, fees are policy-controlled: you may **sponsor** gas for **$0 UX** (including Instant Seal if you choose), **partially sponsor**, or **charge** your own internal gas schedule. Most costs appear on **cross-layer moves** (e.g., settling to L2/Bitcoin) or **managed Spur** services. Optional **Seal Credits** (sats-pegged) help budget per-seal costs. Protocol/DAO splits follow the Economics: Fees, UDS Staking, Anchor Pot & Revenue section (e.g., **7% DAO skim** per epoch).

Capacity & infra. Typical validator (L2 or L3): **~8 vCPU / 32 GB RAM / NVMe**; gateways/indexers scale elastically. Storage is **content-addressed**: BYO (S3/WebDAV/NAS) or managed Spur, with on-chain Usage/Storage Receipts. Gateway egress is **~5–15 Mbps per active 100 users** for static sites.

Commons Live ceilings. Mesh (ultra-private) suits **2–6** participants. **SFU** mode handles **~40–80** participants on **8 vCPU/16 GB** (BYO) and **low-hundreds** per room on Spur (codec-dependent).

QoS & fairness. Priority lanes favor **treasury/payroll/tax** (often with Instant Seal) over background flows. At L2, orgs receive **weighted-fair inclusion** and may temporarily **raise seal cadence** under load.

Health SLOs & graceful degradation. Targets: ≥99.9% anchor-slot hit over 30 days; warn at 5 min late; page at 10 min late; CRL freshness ≤60 s; 100% inclusion of valid seals. Degradation paths: anchoring at lower feerates, storage failover (BYO↔Spur), SFU auto-caps / audio-only fallback.

Determinism & audit. Every block commits (chainID, height, receiptsRoot, moduleRoots[], paramsHash, timeRange). Explorers verify the proof path **Event → L3 BlockRoot → L2 AnchorTree → Bitcoin (Taproot)**.

Defaults & limits. Max tx: 64 KB (L2) / 96 KB (L3-configurable); target ≥90% block fill; Cut-off names ≤64 bytes; blob soft cap 100 MB/CID. Live: mesh 2–6, SFU ~40–80 (BYO) or 100+ (Spur); must include scope-type:identifier; unique within block.

BYO vs Spur. BYO = your infra (VMs/K8s, your SFU capacity). Spur = managed scale with **SLA + credit receipts**. **Switching providers is a URL/config change** via provider-agnostic, signed manifests.

L2 consistently hits 10-minute blocks + ~2 s Instant Seals; anchors land twice daily with visible proofs; L3s deliver sponsored $0 UX or charged gas by policy; Commons Live meets capacity caps with E2E media; BYO↔Spur swaps require no code; every deploy/receipt/anchor resolves along the canonical proof path.

Protocol Architecture & First-Principles Design

RAABX runs two tightly-coupled L2 chains with Bitcoin-style 10-minute blocks and two explicit controls: Instant Seal (close & publish a block now, ~2 s) and Cut-off Seal (freeze a specific business scope without stopping the block). If no Instant Seal occurs, Auto Seal closes on the 10-minute cadence. All important events roll into fixed, twice-daily Taproot anchors on Bitcoin; seals never trigger anchors.

Consensus is a stake-weighted Nakamoto design on a modern BTC fork with VRF leader election. Instant Sealed blocks are final by rule via a SealCert (p95 ≤3 s) (≥2/3 of a sampled, stake-weighted committee); otherwise finality follows the 10-minute longest-chain rule. Proposers must include the earliest valid Instant Seal tx; per account-key rate limits prevent abuse. Violations are slashed by rule. Sample size 512 and a ≥67% sampled-stake threshold are chosen to meet the p95 ≤3 s target while preserving safety.

Instant Seal — Earliest-Valid Determinism & Slashing

Objective queue. All Instant-Seal requests enter an append-only **SealQueue**. Each request gets a monotonic **Eligibility Index (EI)** at enqueue time (0, 1, 2, …). EI is a counter—not wall-clock time. Every block header commits the queue via **SealQueueRoot**.

Validity rule (no clocks). A proposed Instant-Sealed block is valid only if its included transactions are exactly the **ready prefix** of the queue at proposal time, ordered by (EI, txid). "Ready" means: correct nonce, dependencies included, and fee meets the floor.

Slashable condition. If a proposer includes any tx with higher (EI, txid) while **skipping** a ready tx with a lower (EI, txid), that block is **slashable**.

Evidence (fully objective).
(a) the committed SealQueueRoot at proposal start;
(b) Merkle proofs for **the skipped lower-EI tx** and **the included higher-EI tx**;
(c) minimal readiness witnesses (nonce, deps, fee).
No timestamps are needed anywhere.

Example. If tx **A** has EI=120 and is ready, and the block includes tx **B** with EI=123 while omitting **A**, the proposer is slashable.

Provenance commits every business action: ScopeRoot → ModuleRoot → BlockRoot → AnchorTree → Taproot. Anchoring is cost-smoothed and funded by the Anchor Pot (primarily Instant Seal fees); a controller adjusts the Instant Seal floor to maintain runway, with fixed 2×/day anchors and a ≥1/24h fallback.

minSeal Controller — Fee Units & Data Sources

Definition. p95Fee refers to the **Bitcoin L1** fee rate, measured as the 95th-percentile mempool price in **sat/vB**. A BTC Fee Oracle posts the **median of ≥3 sources** plus a timestamp. L2 prices/quotes remain in **RBS/byte**; the controller converts L1 cost into an L2 floor so anchors remain funded even when L1 fees spike. If oracles are stale, a policy floor is used.

Computation (every 30 s).
Inputs

- p95_sat_per_vB: from the BTC fee oracle (fresh ≤120 s)
- P_BTC_per_RBS: RBS/BTC TWAP price oracle (fresh ≤120 s)
- anchor_vbytes: expected Taproot anchor size (default **200 vB**)
- amort: EWMA of expected L2 txs per seal (clamped **[64, 4096]**, min 1)
- base_floor_rbs_per_byte: policy floor on L2 (default **1e-9 RBS/byte**)
- tx_size_bytes: size of the quoted L2 transaction

Formula

```
min_seal_rbs_per_tx =

  max( base_floor_rbs_per_byte * tx_size_bytes ,

      (p95_sat_per_vB * anchor_vbytes / amort) / 1e8 / P_BTC_per_RBS )
```

Behavior on stale data. If either oracle is older than **120 s**, hold the previous min_seal_rbs_per_tx (no jump up or down).

Units sanity. sat/vB → BTC via /1e8, then BTC → RBS via / P_BTC_per_RBS. L2 wallets display fees in **RBS**; the controller ensures anchors remain funded from **Instant Seal** fees.

Security posture ("RAAB-Ω") includes hybrid ECDSA+PQC keys, role-separated keying with time-locked rotation/recovery, QUIC+Noise transport, and an optional encrypted Private-

lane mempool; MEV hygiene uses deterministic ordering within the Private lane plus admission controls and inclusion rules. Economic base layer: UDS auto-delegation (every balance earns by default), VRF-based operator selection weighted by stake^β × health, compounding on by default, and slashing/auto-unbond for poor health.

Data semantics: the Public L2 exposes customer-visible receipts (no PII), while the Private L2 keeps shielded accounting with auditor viewing keys and off-chain encrypted blobs proven by StorageReceipts. Test anchoring uses RAAB-Signet (3-of-5 HSM, ~10-minute blocks). A single Gateway origin serves wallet, RPC/WebSocket, seal quotes, explorers, and health. Threat drills cover seal-spam, key compromise, oracle/CRL staleness, SFU DoS, storage outages, equivocation, and missed anchors, with concrete pass conditions. Initial defaults set block/anchor cadence, adaptive Instant Seal floor, allow-listed Cut-off scopes, hybrid keys, and validator specs; implementation notes define seal handlers, end-of-block commits, the twice-daily anchorer, explorers, and telemetry.

SealCert Signature Aggregation & On-Chain Encoding.

We use BLS12-381 (IETF draft v4 hash-to-curve, Basic scheme with proof-of-possession at validator registration) to aggregate committee signatures into a single signature. For a 512-member committee with threshold q, each signer BLS-signs the same SealCert message and their signatures are aggregated; the aggregated public key is the sum of participant keys selected by a bitmap.

Certificate format:

```
SealCert {
  version:                uint8 // =1
  seal_id:                bytes32 // unique ID for this seal/proposal
  block_hash:             bytes32 // hash of sealed L2 block or state root
  committee_root:         bytes32 // Merkle root of the 512 registered BLS pubkeys
  quorum_threshold_q:     uint16 // e.g., 344 for ~67% of 512 (default)
  participant_bitmap:     bytes64 // 512 bits, little-endian
  agg_pubkey:             bytes48 // BLS12-381 G1 compressed
  agg_signature:          bytes96 // BLS12-381 G2 compressed
  domain:                 bytes32 // DST for domain separation (chain+epoch)
}
```

Verification. Verifiers (and the L2 contract) reconstruct the aggregated public key by

summing the registered keys indicated by `participant_bitmap` (membership proven against `committee_root`), then check a single aggregate BLS signature over `(version, seal_id, block_hash, domain)`. This yields one succinct certificate of ≈339 bytes total (1 + 32 + 32 + 32 + 2 + 64 + 48 + 96 + 32), verified with a single aggregate BLS check. Bitcoin anchors do not verify BLS; they only carry commitment roots.

Security notes. PoP at registration prevents rogue-key attacks; the bitmap ensures objective signer sets; `domain` prevents cross-protocol replay.

Token Supply, Allocation & Rewards

Totals & units. The network token is RBS, hard-capped at 1,104,080,000. The base unit is the raaboshi. For staking math the docs use R-Stake = 100,000 RBS (a display unit only).

Epoch & cadence. An epoch = one sealed block. By default blocks Auto Seal ~every 10 minutes; Instant Seal ~2 s is optional (paid); Cut-off Seals fence a business scope but don't change payout timing. Anchors to Bitcoin happen on a fixed 2×/day schedule and are independent of rewards.

Genesis allocation (100%).

Rewards Reserve 49% — funds emissions over 10 years.

KYC Starter 20% — covers paid, zero-PII onboarding at scale.

DAO Treasury 7% — governance, grants, ops.

Ecosystem & Partnerships 8% — integrations, incentives.

Market Making & Liquidity 5% — exchange/liquidity programs.

PS-10 Program 5% — network public-service operators; includes a base lock (10 × 100k RBS, permanent) and a Founders Delegation Pool (~54.204M RBS) that decays and is health-gated.

Team & Contributors 5% — long-term aligned vesting.

Anchor/Ops Float 1% — ensures anchors clear during fee spikes.

Emissions (from the 49% Rewards Reserve). Pre-minted, time-locked, and streamed per sealed block for 10 years, peaking early then tapering: Y1 18%, Y2 20%, Y3 14%, Y4 12%, Y5 10%, Y6 8%, Y7 6%, Y8 5%, Y9 4%, Y10 3% (total 540,999,200 RBS). Each year has a hard cap; unused for a year does not spill into later years. Per-block emission adjusts dynamically so the yearly cap is met regardless of the number of sealed blocks; unused yearly allocation does not roll over.

What funds each epoch's pool.
RewardPool = Fees + Emission_for_this_block.

Normal tx/service/Router fees flow in directly.

Instant Seal fees first top up the Anchor Pot to maintain a ~30-day anchoring runway; if the runway is already full, any overflow returns to fees (so RewardPool isn't shorted or double-counted).

Payout order (every epoch).

DAO skim: 7% of RewardPool goes to the DAO Treasury.

Perma Pot overlay (PS-10s): from the remaining 93%, route a small fixed share to PS-10s, split equally among online PS-10 operators: Y1 5%, Y2 4%, Y3 3%, Y4 2%, Y5–Y10 1%, then 0% thereafter.

Participatory Pool (the rest): paid out by UDS (Universal Delegation & Staking).

Self-operated stake: keeps 100% of its UDS share.

Delegated stake: 25% to the holder / 75% to the operator.

Operator weighting uses stake^β × health; poor health can auto-unbond, and slashing applies for bad behavior.

Why this works:

Predictable UX & audit: rewards stream at block cadence; anchors are fee-smoothed and guaranteed by the runway.

Bootstrapped public service: PS-10's Perma Pot + base lock + founders' pool keep essential network services online from day one.

Fair participation: everyone earns by default via UDS; strong operators with healthy uptime earn proportionally more.

Cost isolation: Instant Seal fees fund anchoring first, so auditability isn't hostage to market spikes; overflow flows back to users via the RewardPool.

Example. If an epoch's RewardPool = 100 RBS:

7 RBS → DAO.

Remainder 93 RBS → if Year 1, 5% (= 4.65 RBS) → PS-10s (split equally among online PS-10s).

88.35 RBS → Participatory Pool via UDS (self-run keeps all; delegated splits 25/75).

Economics: Fees, UDS Staking, Anchor Pot & Revenue

What fills the rewards each block ("pay period")

Every time the network **closes a block** (about every 10 minutes by default), it creates a pot of rewards for that block (we call a closed block an **epoch**):

RewardPool = Fees collected in this block + the scheduled Emission for this block.

Fees include: ordinary gas/service fees, **Router/DEX** basis-points (as those go live), and any approved module/program fees.

Important exception — Instant Seal fees: before counting as "Fees," they **first refill the Anchor Pot** (see below) so the network can **guarantee two Bitcoin anchors per day for ~30 days**. Only **overflow** (if the Pot is already healthy) flows back into Fees. This prevents double-counting and makes auditability independent of market spikes.

Translation: think of Emission as a planned drip of tokens each block, and Fees as what users paid this block. Instant Seal fees go to the "anchoring budget" first so audit receipts are never underfunded.

Who gets what (split order)

When the **RewardPool** is ready, the network pays it out **in this exact order**:

DAO skim — 7% of RewardPool goes to the DAO Treasury (funds grants, ops, upgrades).

PS-10 overlay (Perma Pot) — from the **remaining 93%**, a **small, fixed share** goes **equally** to **online** PS-10 operators:

Year 1: 5%

Year 2: 4%

Year 3: 3%

Year 4: 2%

Years 5–10: 1%

After Year 10: 0%
In addition, PS-10 has a **permanent base lock** of **10 × 100k RBS** and a **Founders Delegation Pool (~54.204M RBS)** that **decays over time** and is **health-gated** to bootstrap essential public services.

Participatory Pool — the remainder, paid by **UDS** (Universal Delegation & Staking) to operators and stakers (details next).

UDS — how staking pays "by default"

UDS is designed so **everyone earns by default**, while **operators who keep the network healthy** earn more.

Auto-delegate: when RBS lands in your wallet, it **auto-bonds to a healthy operator**; if you transfer it, it **re-bonds at the destination**. No micromanagement needed.

Self-operated stake: if you run a node and meet health SLOs, you keep **100%** of your UDS share for that stake.

Delegated stake: 25% of that stake's UDS share goes to the **holder**, **75%** to the **operator**.

Operator weighting = stake^β × health (uptime, inclusion behavior, responsiveness). If health drops, delegated stake can **auto-unbond** to protect holders. **Slashing** applies for serious faults (e.g., equivocation, inclusion-rule violations).

Compounding: on by default; you can switch to "send to spend" if you want rewards paid out instead of re-bonded.

Translation: if you do nothing, your RBS picks a good operator and earns. If you run a reliable node yourself, you keep everything that stake earns.

Seals, anchors & the Anchor Pot (why anchors are always funded)

Seals control **timing inside RAABX**:

Auto Seal

Instant Seal closes & publishes the current block **now** (~2 s publish) for time-sensitive actions.

Cut-off Seal freezes **one business scope** (e.g., a payroll run) inside an open block so accounting is clean, without stopping other traffic.

Anchors are **separate**: the network posts cryptographic commitments to **Bitcoin exactly 2×/day** on a fixed schedule. **Seals never trigger anchors.**

Anchor Pot is the **budget** that pays Bitcoin fees for those twice-daily anchors.

Instant Seal fees fill the Anchor Pot first to maintain a **~30-day runway**.

If the runway dips, the network **nudges up the Instant Seal floor** (minimum fee) so the Pot refills predictably; when healthy, the floor relaxes.

If the Pot is already full, **overflow from Instant Seal fees** is counted as block Fees and thus enters the RewardPool.

Translation: the system guarantees predictable, on-time Bitcoin receipts by pre-funding them. Urgent "seal now" users effectively co-fund that guarantee.

Fees, prices & UX

Inside your org's L3s: you can **sponsor gas** (even **min gas = 0**) so staff and customers see **$0 UX**. Most costs only appear on **cross-layer** moves (settling out to L2/BTC) or when using **managed services** (e.g., Spur).

Instant Seal floor adapts to network fee pressure and Anchor-Pot runway, keeping **costs predictable** while guaranteeing anchors.

On-chain revenue programs (beyond gas)

BusinessNet Search/Ads — clearly labeled, in-wallet placements. Revenue shares **shift over time**:

Y0–2: 40% Team / 40% DAO / 20% Stakers

Y3–4: 30% Team / 45% DAO / 25% Stakers

Y5+: 20% Team / 50% DAO / 30% Stakers

Listings/marketplaces & template sales — optional takes with **automatic, on-chain split receipts**.

Router/DEX fees — basis-points on matched volume as **cross-org liquidity** goes live.

Why this design works (in practice)

Anchors always funded: Instant Seal fees **top up Anchor Pot first**, so Bitcoin audit receipts are **never hostage** to volatile fees.

Fair by default: UDS makes every RBS earn by default and **rewards operator health**, not just size.

Predictable DAO income: the **7% skim** gives steady resources for grants, ops, and upgrades.

Public services bootstrapped: PS-10 base lock + founders pool + decade overlay ensure critical services are online **from day one**, then **taper**.

Example (Year 1)

Suppose a block's **RewardPool = 1,000 RBS** in **Year 1**:

DAO skim (7%) → **70 RBS** to the DAO.

PS-10 overlay (from remaining 930 RBS)

5% of 930 → **46.5 RBS** split **equally** among **online** PS-10s.

Participatory Pool (remainder)

883.5 RBS → paid out by **UDS**

Self-operated stake: keeps **100%** of its UDS share.

Delegated stake: 25% to holder / **75%** to operator.

Anchor-Pot note: if Instant Seal fees **overflowed** the Anchor Pot this block, that overflow was already included in **Fees** (and thus in the **1,000 RBS**). If the Pot **needed a top-up**, those Instant Seal fees were **used for anchors first** and **did not** reduce the RewardPool (anchors first, then users).

Recap

Each block is a short **pay period**: **Fees + Emission** → **RewardPool**.

Instant Seal fees first secure the **Anchor Pot** (Bitcoin receipts 2×/day), then overflow counts as Fees.

Payout order: **7% DAO**, then **PS-10 overlay** (declining over 10 years), then **UDS** pays the rest to stakers/operators.

UDS means **you earn by default**; healthy operators earn more; self-run keeps 100%; delegated splits **25/75** (holder/operator).

Orgs can give **$0 UX** internally; most costs show up only when crossing layers or using managed services.

Ads/listings/Router add **new revenue** streams with on-chain, auto-split receipts.

Perma-Stake (PS-10): Founding Validators & Perpetual Rights

Offering & Terms moved to Appendix A (protocol paper stays protocol-focused).

What a PS-10 is. A supply-capped set of **10** transferable NFTs with a permanently bonded, non-transferable 100,000 RBS stake. Each PS-10 permanently **bonds 100,000 RBS** to a validator identity and confers "Founding Validator" status. The NFT **is transferable** (ownership and validator rights move with it), while the **100k RBS remains bonded** to the validator slot. You earn **normal validator rewards** forever, plus a decade-long overlay.

Earnings (how cash flows). Every sealed block forms a RewardPool. Distribution order: **(1) DAO 7% skim** → **(2) Perma Pot overlay** to **online PS-10s split equally** → **(3) remainder via UDS** to all stakers (operators/holders).
Perma Pot schedule: **Y1 5%**, **Y2 4%**, **Y3 3%**, **Y4 2%**, **Y5–Y10 1%**, then **0%**.
UDS rules: if you **self-operate** and meet SLOs, you keep **100%** of your share; **delegated stake** splits **75% operator / 25% holder**. Operator weight = **stake^β × health**; poor health can **auto-unbond**, and slashing applies for misbehavior. (Separately, **Instant Seal fees fund anchors first**, so Bitcoin anchoring is always covered; overflow returns to fees.)

Rights & governance. PS-10 holders get a **visible badge** in explorers and a **seat in the Perma-Stake Senate** (oversight on large DAO outflows and anchor-schedule changes). Benefits are tied to **real network usage**; all receipts and anchors show on the canonical proof path to Bitcoin.

Operating model. Run your validator on **BYO hardware** or use **Spur-managed** nodes; switch at will without losing status. Use **HSM/TPM or secure enclaves**, least-privilege networking, and keep keys rotation-ready. Health SLOs must be maintained to receive overlay and delegation flow.

Governance

RAABX governance keeps the network upgradeable without risking funds, privacy, or auditability. It's quadrant-based and simulation-first: proposals are created and reviewed in-wallet, run a required Σ-Bond simulation, then move to voting with stake snapshots, delegations, visible timelocks, and optional Instant Seal for fast finalization (anchors stay on their fixed Bitcoin cadence). Validators get release-gate and emergency controls in node TUI/CLI; explorers show a full Draft→Execute→Anchor timeline.

Decision-making is split across four "seats." Token-holder Assembly (all staked RBS) decides economic and user-visible policy. A 9-member Policy Council steers human-policy surfaces (KYC/KYB lists, sanctions sources, ranking defaults, risk limits). The Validator Committee (all healthy validators) guards technical safety and can veto on liveness/safety risk. The Perma-Stake Senate (10 PS-10 holders) oversees capital-at-risk, anchors, and large

treasury moves, with time-boxed treasury/anchor veto and co-sign for direct anchors. Together they ensure users/capital, policy stewards, operators, and founding infra agree on meaningful changes.

Seat badges are SBNFTs so wallets/nodes can gate actions and create clear audit trails: PCSeat (term-limited), VCSeat (epoch-scoped), PS-10 NFT (defines Senate seat), and optional DAO-Signer. TA voters don't need a badge (stake snapshots handle that).

Constitutional guardrails lock the foundation: total supply hard cap, add-only crypto, no PII on-chain, fixed anchoring cadence, L3 sovereignty, immutable emission locks, and seals that can't trigger out-of-band anchors; amending these needs an 80% TA supermajority plus VC and PSS at ⅔ and PC majority, then a 60-day timelock with reopenable veto. Day-to-day changes flow through wallet-native classes with set paths, quorums, and locks—ParamChange, TreasurySpend, ReleaseGate, AnchorSchedule, CryptoAdd, PolicySet, ProgramSplit—and all require a Σ-Bond pass.

Before any vote opens, the Σ-Bond check must show bounded harm (<1% 30-day loss per cohort), no liveness or privacy regressions, and adequate anchor-runway; failing sims block the vote. Lifecycle: Draft→Open Review (≥72h)→Σ-Bond PASS→Vote→Timelock (with seat-gated veto)→Execute→Anchor→Post-mortem. Emergency levers are narrow/self-expiring: Pause Module (VC⅔+PC majority, ≤72h, TA ratify to persist), Direct Anchor Now (PSS 6/10 + VC simple), and temporary Instant Seal floor raise (auto-expires 72h, TA ratify to persist). Initial parameter envelopes (e.g., staker share 80–95% of RewardPool; router fee 0–30 bps; 2×/day anchors ±30m) and minimal on-chain data types (Proposal, Vote, SimReport, GovernanceReceipt, AnchorEmergencyReceipt, seat badges) are defined.

Infrastructure & Operations

Operator models. Three ways to run: **BYO** (your infra), **Spur-managed** (SLA with credits), or **Hybrid**. Dual-pin cutover and signed, content-addressed **Site Manifests** let you switch providers/regions safely with no trust leap.

Service classes & SLOs. Clear targets per class: **Core (seals/consensus) ~2 s Instant Seal**, **2 anchors/day** with small jitter; **Gateway/API**, **Storage retrieval**, **SFU (real-time media)**, **DevPods** each have published SLOs/quotas. SLA breaches auto-issue minute-credits to affected org wallets.

Roles & sizing. Gateways, validator quorum, anchoring aggregators (with hot standby), dual-pinned storage, SFU capacity, and oracles are sized explicitly. **Keys are hardware-backed** (HSM/TPM or enclaves) with scheduled and emergency rotation (time-lock + old-key veto). **Cluster traffic** uses **QUIC+Noise**; **edge** is **TLS 1.3**. Strict allow-lists and rate-limits everywhere.

Durability & anchoring. State snapshots, receipts, and Anchor-Pot data are **content-addressed** and geo-replicated (often BYO+Spur). Normal anchoring is **2×/day**; the aggregator maintains fees/runway. Narrow, fully-logged emergencies exist (e.g., **Direct Anchor**, temporary Instant Seal floor raise).

Deploys & cutovers. Blue/green deploys gated by health; domains bind via /.well-known/raabx-site to a signed manifest. **Atomic rollbacks** by DeployID. DevPods have per-org quotas and exported telemetry.

Runbooks (incidents). Playbooks define actions for: missed anchor slots, validator equivocation/double-sign, storage outages, SFU DoS or capacity caps, KYC-oracle downtime, suspected key compromise, Instant Seal spam/runway risk—each with immediate steps, any required council approvals, and receipts to publish after.

Provider posture. Spur providers onboard with a public manifest and locked **collateral**; synthetic probes + customer receipts measure SLOs. Breaches drain the collateral vault and can **auto-throttle** service until topped up. BYO operators publish SLO endpoints but post no collateral.

Observability & security. Standardized **OpenTelemetry** with dashboards for core metrics (e.g., block_close_age, seal_latency_p95, anchor_pot_balance). Wallet shows next-anchor countdown and last Bitcoin txid. **Signed releases** with enforceable release-gates; **≤72h** patch window for critical CVEs; **data-residency tags**; **no PII on-chain**; exportable audit packs.

Ops acceptance (what "good" looks like). Instant Seal **~2 s**, dependable **Cut-off** seals, both daily anchor slots hit (±15m smoothing), clean failovers BYO↔Spur, and successful drills for all emergency paths—each with verifiable receipts and the canonical proof path to Bitcoin.

Compliance & Legal Posture

Purpose & stance. Keep **PII off-chain** and **keys with users**; make duties explicit; prove integrity via **Bitcoin anchors**; run all compliance actions **in-wallet/in-node**. Principles: **data-minimization**, **separation of duties** (Foundation ≠ Spur ≠ each org), **crypto-agile add-only** algorithms, **audit-by-design**, and **predictable provenance** (Auto/Instant/Cut-off Seals; **anchors 2×/day** on a fixed schedule).

Who does what (RACI).

Foundation (protocol steward): L2/L3 software, seals logic, anchor rules, wallet/node UX; **no custody**.

Spur (managed provider, optional): storage/compute/SFU under SLA; **processor**, not custodian.

Organizations (Controllers): their L3 policy, data, contracts, DSARs, consent, legal holds.

KYC Oracles (issuers): off-chain ID/sanctions; revocation lists on-chain.

Validators: security, availability; no PII; accept slashing/IR duties.

Users: hold keys, consent, backups.

KYC/AML, sanctions, Travel Rule. Zero-PII on-chain. Users verify with **two independent oracles**; wallet attaches fresh **"valid & unrevoked"** ZK proofs with **revocation freshness** enforced. Travel-Rule payloads (if used) stay **off-chain** between regulated counterparties.

Money transmission & custody. Wallets are **non-custodial**; fee sponsorship ≠ custody. Spur operates services, **not keys**. On/Off-ramps are licensed connectors; custody sits with the PSP. Protocol **routes & proves**, seals control timing; anchors stay fixed-cadence.

Token law posture. RBS = hard-capped fee/staking utility; rewards are **fees + scheduled emissions** (no "profit promises"). **PS-10** offered to **qualified purchasers** with **geofencing**; eligibility checks run in-wallet; counsel drives jurisdictional paths.

Privacy & data protection (GDPR/CCPA-ready).

Controller/Processor mapping: org = Controller; Spur = Processor; Foundation/validators handle **pseudonymous receipts only**.

Residency: every **StorageReceipt** carries a region tag; gateways enforce in-region store/fetch.

Right to erasure/DSAR: delete/rotate ciphertext at provider; on-chain **hash receipts** remain as non-PII integrity records; DSAR export flows ship from Dev Studio.

SOC2/ISO: managed providers publish attestations; **AuditReceipts** anchor.

Content & comms (Commons/Mail/Live). AUP blocks illegal/abusive content. **E2EE** for private rooms/DMs/Mail; moderators act at the **envelope** level (rate/ban), not content. Recording shows **consent banners**; legal-hold snapshots are opt-in and logged.

Ads, ranking & disclosures (BusinessNet). Sponsored items are **labeled** and mint **AdReceipts**. Ranking uses **non-PII** signals (uptime, freshness, disputes, anchor age).

Tax & reporting. OpenTax templates (VAT/GST/sales/payroll/excise). **Cut-off Seals** give clean period boundaries (payroll runs, Z-reports) with explorer visibility.

Law enforcement & legal holds. RAABX can't produce plaintext it doesn't have. Normal path: verify order → notify org (Controller) → place **LegalHoldReceipt** on encrypted blobs

→ Controller handles disclosure off-chain. Emergency timestamping can use a narrowly-scoped, seat-gated path.

Incident response. Triage ≤1 h; public note ≤24 h; regulatory windows (e.g., GDPR 72 h). Key compromise: pause narrowly, **rotate keys with time-lock**, publish **Incident/Fix Receipts**. All IR happens in-wallet by authorized roles.

Contracts (BYO vs Spur).

BYO: policy templates for security/logging/audit; org proves SLOs in explorers.

Spur: MSA + DPA + **on-chain SLA credits** from provider collateral; right-to-audit, pen-tests, portability. Wallet shows provider SLOs + credit receipts.

Jurisdictions. PS-10 and certain flows are **geofenced**; embargoed/sanctioned regions are blocked at KYC/payment. Dev Studio warns on template/regulatory mismatches.

Checklists (in-wallet).

Orgs: name DPO, publish privacy policy, sign DPA (if Spur), set residency tags & Cut-off presets, enable consent/legal-hold, vet licensed ramps, train staff on keys/E2EE, define DSAR.

Validators: HSM/TPM, quarterly rotation, hardening; publish contact/jurisdiction; keep node at ReleaseGate; accept slashing/IR.

Spur: maintain collateral, publish SLOs, pass SOC2/ISO, keep DPAs/SCCs current, pen-test & breach comms cadence.

Acceptance (what "good" looks like). No **PII on-chain**; KYC revocation freshness enforced; seals produce receipts; **anchors land on schedule**; SLA credits/incidents/audits/legal-holds emit on-chain receipts; ads labeled with on-chain splits; BYO/Spur DPAs & right-to-audit in place; **region flags enforced** at gateways. Net effect: **narrow regulatory surface by default**, with provable, Bitcoin-anchored audit trails.

Launch Plan & Acceptance

Phases (what ships, in order):

Devnet: spec freeze, seal/anchor plumbing, RAAB-Signet anchors, faucets, wallets + explorers, Σ-Bond simulation harness.

Public Testnet: zk-KYC (two-oracle), Passports, Commons (chat/DM/Live), Mail, StorageReceipts, Auto/Instant/Cut-off Seals end-to-end, basic OpenTax packs, Anchor Pot controller running with twice-daily Taproot posts.

BusinessNet Alpha: Pages hosting + Deploy Manifests, BYO/Spur toggles, in-wallet domain binding, search/ranking with labeled ads and on-chain AdReceipts.

Launchpad L3 Beta: one-click org L3s (Public + Private), Split-Graph (tax/COGS/royalties/payroll accrual), POS + inventory + AR/AP + contracts, payroll runs with scoped Cut-off, $0-UX via sponsored gas.

Router Pilot: cross-org XSwap (HTLC + RFQ), XTender (sealed bids + AwardEscrow), XEscrow (milestones/SLA).

Mainnet Cutover: parameters locked, emissions start, PS-10 overlay live, UDS rewards on, governance quorum online.

Release gates (must pass before advancing).

Safety: Σ-Bond shows bounded harm (<1%/30-day per cohort), no liveness/privacy regressions; fuzz/load soak (p95 seal latency, block fill targets), CRL freshness bound holds.

Provenance: canonical proof path verified (Event → L3 BlockRoot → L2 AnchorTree → Bitcoin) with explorer drill-downs.

Economics: Anchor Pot runway ≥30 days; RewardPool accounting (DAO skim, PS-10 overlay, UDS splits) matches ledger; Instant Seal floor adapts as designed.

Ops: blue/green deploy with rollback, dual-pin storage retrieval, BYO↔Spur cutover, incident runbooks exercised (missed anchor, equivocation, storage/SFU outage, oracle lag).

Governance: ReleaseGate vote + timelock; emergency levers proven to auto-expire.

Acceptance (What each stage should look like when completed):

Testnet: twice-daily anchors hit (±15m), Instant Seal ~2 s publish, Cut-off seals fence scopes without closing blocks, zk-KYC revocations enforce within freshness window, explorers show Merkle paths.

BusinessNet Alpha: deploys are signed; origin bound via well-known; assets fetch with StorageReceipts; ad placements labeled with on-chain splits; downtime credits mint automatically on breach.

L3 Beta: retail demo ("Candy Bar") posts Split-Graph correctly; payroll demo accrues cents-per-sale and pays with scoped Cut-off; refunds/voids reverse postings; all anchor to Bitcoin on schedule.

Router Pilot: atomic L3↔L3 swaps succeed or refund; tenders commit-reveal with deterministic awards or committee mode; escrows release/rollback per policy.

Mainnet: emissions stream per block, UDS auto-delegation active, PS-10 overlay pays only to online slots, DAO skim visible, governance motions flow Draft→Execute→Anchor.

Data, migration, and fallbacks.
Content is content-addressed; snapshots/export packs allow replay to any height; emergency **Direct Anchor** (policy-gated) exists but routine anchors stay fixed 2×/day. BYO and Spur remain interchangeable via signed manifests; provider swap is a config change, not a redeploy.

Bounty & audit program.
Public bug bounty (consensus, seals, Anchor Pot controller, zk-KYC circuits, Router), third-party audits on consensus/seal handlers/Router/zk-verify, and post-mortems anchored on-chain for any sev-rated incident.

Go-live checklist (condensed).
Keys in HSM/TPM; domain bound; OpenTax packs loaded; Anchor Pot runway green; ReleaseGate passed; runbooks drilled; governance seats seated; PS-10 validators online; wallet shows next-anchor countdown and last Bitcoin txid for all surfaces.

Risks, Limits & Safeguards

This section maps RAABX's full risk surface and the built-in mitigations across governance, technical/crypto, operations, markets/economics, compliance/legal, security/abuse, and Bitcoin-anchor dependencies, then clarifies what RAABX can and cannot prove.

Technical & cryptographic. Possible issues include consensus/state-machine bugs, ZK verification faults, PQC fragility, MEV/ordering and mempool privacy leakage, anchoring pipeline misses, and off-chain data availability gaps. Mitigations include staged rollouts with ReleaseGate and audits, hybrid PQC with time-locked rotations, inclusion rules + monitoring/penalties, threshold Taproot keys with smoothing/runway SLO and emergency "Direct Anchor Now," and dual-pin storage with retention, drills and integrity checks.

Operational & organizational. Risks span key/ops hygiene, provider (Spur/BYO) outages, and human-factor frictions (zk-KYC, naming, wallet learning, fee sponsorship). Controls include HSM/TPM + two-person rules and rotations, sentry topology/no-SSH, on-chain collateralized SLAs with auto-credits, failover + steady snapshots/restore drills, and in-wallet training/status plus preset fee grants.

Economic & governance. Token/fee volatility and thin liquidity (post-MVP Router) can stress Instant Seal economics; program revenues may underperform. Mitigations: adaptive

minSeal and smoothing bands, stable-unit quotes, phased Router with allow-listed pairs, circuit breakers/HTLC fallback, diversified fees and transparent dashboards. Governance risks (gridlock/capture, emergency power misuse, parameter drift) are checked by multi-body approvals, timelocks and Σ-Bond PASS, quorum thresholds, narrowly scoped short-expiry powers with public notes, constitutional envelopes, simulation/diff previews and rollback levers.

Compliance & legal. Reclassification, extraterritorial rules (e.g., travel-rule), and privacy law shifts (hashes as personal data), plus payroll/tax correctness and PS-10 sales/secondary trading, all add risk. Responses: zero-PII on-chain with optional TRP payloads between VASPs, residency tags, DPA/SCC frameworks and dashboards; versioned OpenTax packs with hashes, overrides, audits and "simulation before file"; geo/KYC gating, on-chain Founding Validator Agreement, milestone escrows and counsel-validated paths.

Security & abuse scenarios. Threats include phishing/endpoint malware, supply-chain compromises, spam/DoS, and eclipse/censorship; defenses cover OS keychains/passkeys, spend limits and simulation UX, Sigstore/Cosign + SBOMs + pinned CIDs with release-transparency logs, rate/per-key limits and autoscale fallbacks, diverse peers/regions and proposer health + penalties/slashing.

Bitcoin & roadmap. If Bitcoin fees/policy shift, anchors may become costly or delayed; RAABX counters with fee smoothing, 24-hour fallback, dynamic minSeal, governance-bounded retiming and anchor compression.

What RAABX proves (and not). It proves ordering, inclusion and integrity of receipts, who signed/approved, when scopes/batches were sealed, and that these facts were committed to Bitcoin at/after a time. It cannot guarantee third-party plaintext availability, fiat settlement, or correctness of off-chain business judgments—those remain organizational duties.

Run-time guardrails. In-wallet/node early-warning gauges flag low Anchor Pot runway/slot jitter, stale CRLs, slow/failing storage, missed seal-inclusion floors, declining governance participation and rising ZK-verify failures. Shared responsibilities split across protocol/Spur/customer for on-chain safety, off-chain blobs, keys, compliance and security. Features advance devnet→testnet→mainnet only after public SLOs, adversarial/fuzz tests and audits are green, governance PASS with parameter envelopes and rollback plan, ops drills (e.g., snapshot restore, provider failover, Direct Anchor Now), and updated compliance packs/templates are pinned.

Legal, Proof Path & Glossary (Plain-English)

RAABX clarifies that testnet tokens have no value, while the mainnet token (RBS) is a network-utility token used for fees, staking, governance deposits, and Anchor Pot contributions; this paper is not an offer or investment advice. Token use is subject to local laws; operators/users are responsible for compliance. Exact distribution and lockups are defined separately in Distribution Tables.

The "proof path" shows how any action is verifiable on Bitcoin: a business scope closes to a ScopeRoot, scopes roll up into a ModuleRoot, the block commits a BlockRoot, recent BlockRoots aggregate into an AnchorTree, and that root anchors to Bitcoin on a fixed twice-daily schedule. Seals never trigger anchors: Auto Seal closes blocks ~every 10 minutes, Instant Seal closes/publishes a block immediately (final by SealCert), and Cut-off Seal freezes a single business scope inside an open block for clean accounting.

Key terms at a glance: Anchor Pot funds anchors (primarily via Instant Seal fees); BYO is bring-your-own storage; Spur is an optional managed infra/SFU; SBNFT is the non-transferable Passport token; CRL revocation lists keep zk-KYC fresh; UDS auto-delegates stake by health; ReleaseGate defines time/usage unlocks; Σ-Bond is the staking/bonding construct; IRaab20/721 are RAABX token/NFT interfaces; RAAB-Signet is the Bitcoin signet used for end-to-end test anchoring. (Off-chain data durability is reinforced by StorageReceipts, redundancy, client-encrypted backups, and checksums.)

Roadmap & Acceptance Gates

This roadmap is sequence-based (no dates/budgets) and **wallet-first**: every step runs inside the wallet/node, and every artifact—deploy manifests, receipts, reports—anchors to Bitcoin on a **fixed twice-daily cadence**. We only "flip the switch" when all gauges are green: Σ-Bond simulations **PASS**, release gates are satisfied, and each artifact has a verifiable proof path (**Event → L3 BlockRoot → L2 AnchorTree → Bitcoin**). Timing is predictable: **Auto Seal** (~10 min) by default, **Instant Seal** for ~now commits, **Cut-off** to fence scopes; seals never trigger anchors. The design constraints stay constant: two-lane L2 backbone (Public + Private), per-org twin L3s with zero-PII semantics, deterministic auditability, and governance that's simulation- and timelock-gated.

M0 — Private Devnet

Ships: dual L2s (Public/Private), BlockRoot/AnchorTree plumbing, Instant Seal, Anchor Aggregator (RAAB-Signet), hybrid PQC keys, Gateway, early explorers.

Exit gates: p95 commit **~2 s** for 7 straight days; earliest valid Instant Seal always included; **0 missed anchors** across 14 slots; signed snapshot restore **<30 min**; invariants/fuzz pass for receipts, Anchor-Pot accounting, seal economics.

M1 — Devnet Public

Ships: status pane, Σ-Bond simulation harness, faucet UI, metrics dashboards.

Exit gates: Anchor-Pot runway **≥30 days** (adaptive minSeal working); anchor proof pages published **≤5 min** after each slot; two ParamChange dry-runs **PASS** and **revert cleanly**.

M2 — Testnet-1 (Commons)

Ships: Commons (public/org rooms, private/E2E DMs), RAABX Mail, Commons Live (P2P + SFU), StorageReceipts, moderation events.

Exit gates: ChatReceipts for public posts anchor in the **next slot window**; private rooms/DMs verified **E2E** with **no plaintext at rest**; Mail receipts logged; SFU rooms **≥50 participants**, join p95 **≤5s**, optional encrypted recording → StorageReceipt.

M3 — BusinessNet Alpha

Ships: Pages hosting + signed Deploy Manifests, BYO/Spur toggles, in-wallet domain binding, search/ranking with labeled ads and on-chain AdReceipts.

Exit gates: releases **signed**; origin bound via **well-known** fingerprint; all assets fetch via **StorageReceipts**; ads clearly **labeled** with on-chain splits; downtime breaches auto-mint SLA credit receipts.

M4 — Launchpad L3 Beta

Ships: one-click org L3s (Public + Private), Split-Graph (tax/COGS/royalties/payroll accrual), POS + inventory + AR/AP + contracts, payroll runs with scoped Cut-off, sponsored gas ($0 UX).

Exit gates: retail "Candy Bar" demo posts correct Split-Graph; payroll accrues cents-per-sale and pays with **Cut-off**; refunds/voids reverse postings; all proofs anchor on schedule.

M5 — Router Pilot

Ships: XSwap (HTLC + RFQ), XTender (sealed bids + AwardEscrow), XEscrow (milestones/SLA).

Exit gates: atomic L3↔L3 swaps **succeed or refund**; tenders commit-reveal with deterministic or committee awards; escrows **release/rollback** per policy.

M6 — Mainnet Cutover

Ships: locked parameters, emissions **start**, PS-10 overlay **live**, UDS rewards **on**, governance quorum **online**.

Exit gates: emissions stream per block; UDS **auto-delegation** active; PS-10 overlay pays **only** to online slots; DAO 7% skim visible; governance motions flow **Draft → Execute → Anchor**.

Always-on gates (apply to every milestone): every artifact has a **canonical proof path to Bitcoin** (2×/day anchors); bug bounty + third-party audits are active; ops drills (snapshot restore, provider failover, Direct-Anchor) pass; governance ReleaseGate approvals include timelocks; Anchor-Pot runway stays green.

Conclusion

RAABX couples Bitcoin's 10-minute heartbeat with explicit seals that let businesses control settlement timing without sacrificing auditability. All critical events roll into fixed twice-daily Taproot anchors, and Instant Sealed blocks finalize by rule via a sampled SealCert. Identity stays private (zk-KYC; no PII on-chain), balances always earn via UDS, and security is crypto-agile under an add-only governance model. The result is a backbone that operators can reason about—timing is explicit, proofs are portable, parameters are published—and that teams can run today with clear acceptance gates and an upgrade path that never breaks receipts.

Appendix A — PS-10 Offering & Terms

Sale & issuance (primary). Target price: $350k per PS-10 (10 units = $3.5M) to fund ~18–24 months for a 6–12 person team. Funds sit in on-chain escrow, released by milestones: 40% at close, then Devnet public, Testnet (Commons live), BusinessNet alpha, Launchpad L3 beta, Mainnet cutover. Hardware ships ≤30 days; Spur validator ≤5 days. Secondary trading opens after Testnet. Optional sealed-bid with $300k floor and uniform clearing if demand >10.

Investor protections & compliance. On-chain milestone gates with monthly public updates; Senate veto on outsized DAO outflows and anchor-schedule changes; clawback/standstill if two milestones slip >60 days. Offered to qualified purchasers; yields depend on network usage and operator performance; rights governed by the Founding Validator Agreement and on-chain rules.

Lifecycle summary. Year 1–10: overlay pays on top of normal validator rewards; post-Year 10 the overlay ends, but validator yields and the Senate seat continue. Why ten? It seeds credible decentralization while keeping each slot economically meaningful and accountable.

Parameter	Initial	Envelope / Notes
Block target (Auto Seal)	10 minutes	6–12 min via governance
Instant Seal publish time	~2 s	SLO p95 ≤ 3 s
SealCert sample size	512 voters	256–1024 stake-weighted
SealCert threshold	≥67% sampled stake	60–80%
Per account-key Instant Seal rate-limit	1 / 30 s	10–120 s
minSeal floor	max(Fmin, α·p95Fee)·(1+δ) *See "minSeal Controller — Fee Units & Data Sources" for L1 sat/vB vs. L2 RBS/byte units, oracle freshness (≤120 s), conversion, and stale-data fallback*	α=1.0; Fmin=median(p95,12 h); δ=0/0.25/0.5 by runway
Anchor cadence	2×/day UTC	±15 min smoothing
Anchor-Pot runway target	~30 days	warn <30 d; page <14 d

Symbols & Abbreviations (Quick Reference)

- RBS — network token (raaboshi = base unit)
- UDS — Universal auto-Delegation & Staking (always-earning balances)
- Anchor Pot — budget for twice-daily Bitcoin anchors (primarily from Instant Seal fees)
- Perma Pot — decade-long overlay payout to PS-10 operators
- Auto Seal — default block close on a 10-minute cadence
- Instant Seal — close & publish current block in ~2 s; final by SealCert
- Cut-off Seal — freeze a named business scope inside an open block
- SealCert — sampled-committee finality certificate for Instant Sealed blocks
- p95Fee, Fmin, α, δ — parameters for adaptive Instant Seal floor
- ScopeRoot / ModuleRoot / BlockRoot — provenance commitments
- PS-10 — Perma-Stake founding validator slots (10 total)
- Σ-Bond — simulation/bounding gate for governance proposals